# Midterm 2 Practice Problems

MATH 4400/6400 Fall 2026

Give complete justifications, including the relevant hypotheses when applying a theorem. You may quote results proved in class unless a proof is requested.

## I. The division algorithm in quadratic rings

### 1. The division algorithm in $\mathbb{Z}[\sqrt{-2}]$

Find $\gamma, \delta \in \mathbb{Z}[\sqrt{-2}]$ such that

$$5 + 4\sqrt{-2} = (3 + \sqrt{-2})\gamma + \delta, \qquad N(\delta) < N(3 + \sqrt{-2}).$$

### 2. The Euclidean algorithm in $\mathbb{Z}[i]$

(a) Use the Euclidean algorithm in $\mathbb{Z}[i]$ to compute $\gcd(7 + 5i,\, 3 + 7i)$ up to multiplication by a unit.

(b) Express your gcd as an $\mathbb{Z}[i]$-linear combination of the two given elements.

## II. Norms, divisibility, and prime elements

### 3. A rational prime in $\mathbb{Z}[\sqrt{-2}]$

Use the norm to prove that 5 is a prime element of $\mathbb{Z}[\sqrt{-2}]$.

### 4. Norm for $\mathbb{Z}[\sqrt{d}]$

Let $d$ be a nonsquare integer and let $\alpha = x + y\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$ be a nonzero element with $\gcd(x, y) = g$. Prove that $g^2 \mid N(\alpha)$ and that for every $m \in \mathbb{Z}$, $\alpha \mid m \iff (N(\alpha)/g) \mid m$.

## III. Units and prime factorization

### 5. Units in $\mathbb{Z}[\sqrt{2}]$

Show that $7 + 5\sqrt{2}$ is a unit of $\mathbb{Z}[\sqrt{2}]$ and compute its inverse.

### 6. Prime factorization in $\mathbb{Z}[\sqrt{2}]$

Find a prime factorization of $68 + 68\sqrt{2}$ in $\mathbb{Z}[\sqrt{2}]$ and justify your answer.

### 7. A failure of unique factorization in $\mathbb{Z}[\sqrt{-6}]$

Find two essentially different factorizations of 10 into primes in $\mathbb{Z}[\sqrt{-6}]$.

### 8. A failure of unique factorization in $\mathbb{Z}[\sqrt{15}]$

(a) Prove that $2, 3 \in \mathbb{Z}[\sqrt{15}]$ are primes.

(b) Prove that 6 admits two essentially different factorizations into primes in $\mathbb{Z}[\sqrt{15}]$.

## IV. Continued fractions and Pell's equations

### 9. Continued fractions and a Pell equation

(a) Compute the simple continued fraction expansion of $\sqrt{6}$.

(b) Use part (a) to find the fundamental solution $(x, y)$ of $x^2 - 6y^2 = 1$.

(c) Starting from the fundamental solution, construct two further positive integer solutions.

(d) Does $x^2 - 6y^2 = -1$ has any integer solutions $(x, y)$?

### 10. A negative Pell equation

(a) Find the simple continued fraction expansion of $\sqrt{10}$.

(b) Use part (a) to find the fundamental solution $(x, y)$ of $x^2 - 10y^2 = -1$.

(c) Use part (b) to find the fundamental solution $(x, y)$ of $x^2 - 10y^2 = 1$.

(d) Find all units in $\mathbb{Z}[\sqrt{10}]$.

## V. Diophantine equations

### 11. A variant of the Pythagorean equation

Find all integer solutions $(x, y)$ to $x^2 + 2y^2 = z^2$.

### 12. Squares and cubes

Find all integer solutions $(x, y)$ to $x^2 + 1 = y^3$.

## VI. The infinitude of primes

### 14. Primes modulo 6

(a) We proved in class that there are infinitely many primes $p \equiv 3 \pmod 4$. Adapt the proof to show that there are infinitely many primes $p \equiv 5 \pmod 6$.

(b) Suppose that $p \neq 3$ is an odd prime. When is $-3$ R $p$ and when is $-3$ N $p$?

(c) Show that if $p \equiv 5 \pmod 6$ and $x^2 + 3y^2 \equiv 0 \pmod p$ is solvable, then $p \mid x$ and $p \mid y$.

(d) Adapt our proof of the infinitude of primes $p \equiv 5 \pmod 8$ to show that there are infinitely many primes $p \equiv 7 \pmod{12}$. In particular, there are infinitely many primes $p \equiv 1 \pmod 6$.